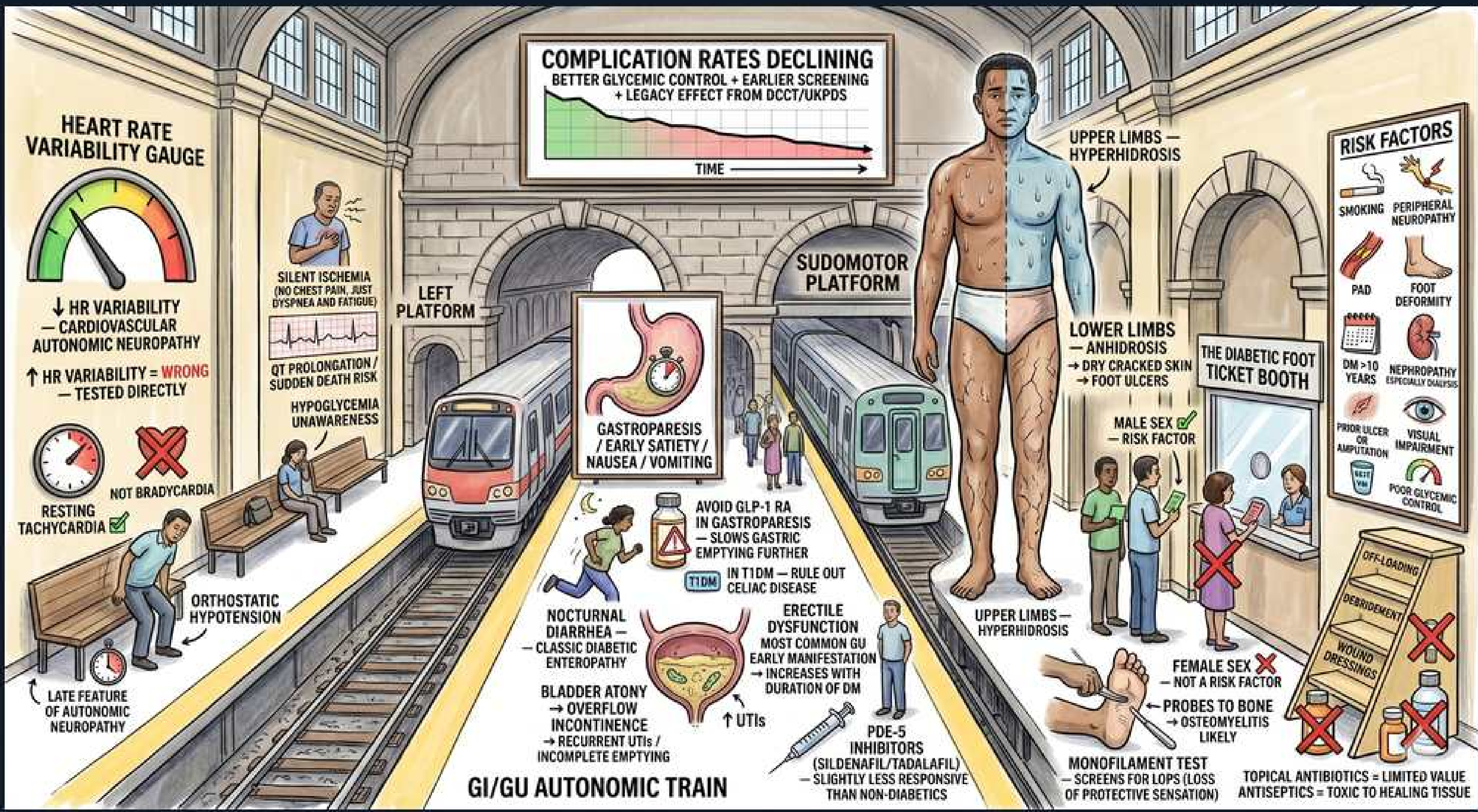
Diabetes — Complication 2: Autonomic & Peripheral Neuropathy / Foot



1. Complication rates declining

WHAT YOU SEE

Banner: complication rates declining with better glycemic control, earlier screening, DCCT/UKPDS legacy effect.

WHAT IT ENCODES

Tight early glycemic control produces a durable 'legacy effect' (metabolic memory) that keeps microvascular complications low for years even after control later loosens. DCCT/EDIC (T1DM) and UKPDS (T2DM) both showed that early intensive control (A1c ~7%) cut retinopathy, nephropathy, and neuropathy long-term. Population complication rates have fallen with earlier screening and better risk-factor control. Mnemonic: 'control early, coast late.'

BOARD TRAP

WRONG to think achieving tight control LATE reverses established complications as well as early control does — the discriminator is timing: the legacy/memory benefit comes from EARLY control, and late intensification (ACCORD) in long-standing T2DM gave no macrovascular benefit and increased mortality with A1c <6%.

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2. Heart rate variability gauge

WHAT YOU SEE

HR variability gauge: decreased HRV = cardiovascular autonomic neuropathy, tested directly.

WHAT IT ENCODES

Reduced beat-to-beat heart rate variability is the EARLIEST objective sign of cardiovascular autonomic neuropathy (CAN), reflecting vagal (parasympathetic) denervation before sympathetic loss. Tested with the expiration:inspiration ratio, Valsalva ratio, and 30:15 standing ratio. CAN affects up to ~60% of long-standing diabetics and independently predicts mortality. Mnemonic: 'vagus goes first, so variability falls first.'

BOARD TRAP

INCREASED heart rate variability is the WRONG answer — autonomic neuropathy REDUCES variability. The discriminator: loss of respiratory sinus arrhythmia (flat HR with deep breathing) is the classic bedside clue to CAN, not an exaggerated HR swing.

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3. Silent ischemia / sudden death

WHAT YOU SEE

Silent ischemia (no chest pain), sudden death risk, QT prolongation, hypoglycemia unawareness.

WHAT IT ENCODES

Cardiac autonomic neuropathy blunts anginal afferents → SILENT myocardial ischemia/infarction, prolonged QT (arrhythmia/torsades risk), perioperative instability, and sudden death. It also causes hypoglycemia unawareness by impairing the autonomic counterregulatory (adrenergic) warning response. Have a LOW threshold to work up diabetics for CAD with atypical or absent symptoms.

BOARD TRAP

WRONG to exclude MI in a diabetic because there is no chest pain — diabetics frequently have painless/atypical MI (dyspnea, nausea, epigastric or shoulder pain). Discriminator: get an ECG/troponin on any diabetic with unexplained dyspnea, CHF, or malaise rather than reassuring them.

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4. Resting tachycardia, not bradycardia

WHAT YOU SEE

Resting tachycardia; NOT bradycardia.

WHAT IT ENCODES

Early CAN damages the vagus first, leaving relatively unopposed sympathetic tone → a fixed resting tachycardia (often HR 90–100) that does not vary with breathing or posture. As denervation advances, the heart rate becomes fixed/'denervated' and unresponsive to exercise or atropine. Mnemonic: 'vagal loss = fast and fixed.'

BOARD TRAP

BRADYCARDIA is the WRONG expectation in autonomic neuropathy — the classic finding is a resting TACHYCARDIA with a fixed, non-varying rate. Discriminator: a fixed HR that fails to slow with deep breathing or rise normally with standing points to autonomic denervation, not a sinus node or conduction problem.

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5. Orthostatic hypotension — late

WHAT YOU SEE

Orthostatic hypotension; late feature of autonomic neuropathy.

WHAT IT ENCODES

Orthostatic hypotension (fall ≥20 mmHg systolic or ≥10 mmHg diastolic within 3 min of standing) is a LATE manifestation of CAN, due to impaired sympathetic vasoconstriction without the expected compensatory tachycardia. Manage non-pharmacologically first (slow position changes, compression stockings, salt/fluid), then midodrine or fludrocortisone if needed. Mnemonic: 'BP drops but pulse won't climb.'

BOARD TRAP

WRONG to call orthostatic hypotension an EARLY sign of autonomic neuropathy — reduced HR variability is early; orthostasis is LATE. Discriminator in diabetic orthostasis: the heart rate does NOT rise appropriately when BP falls (unlike volume depletion, where reflex tachycardia is intact).

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6. Gastroparesis

WHAT YOU SEE

Stomach: gastroparesis — early satiety, nausea/vomiting; avoid GLP-1 RA in gastroparesis.

WHAT IT ENCODES

Vagal GI autonomic neuropathy → gastroparesis: delayed gastric emptying causing early satiety, postprandial nausea/vomiting, bloating, and erratic glucose (insulin acts before food is absorbed → unexpected hypoglycemia then late hyperglycemia). Diagnose with gastric emptying scintigraphy after excluding mechanical obstruction. Treat with small low-fat/low-fiber meals and prokinetics: metoclopramide (watch tardive dyskinesia) or erythromycin (motilin agonist).

BOARD TRAP

GLP-1 receptor agonists (and pramlintide) are the WRONG drug class in gastroparesis — they FURTHER slow gastric emptying and worsen symptoms. Discriminator: choose a prokinetic (metoclopramide/erythromycin) and STOP the GLP-1 RA, rather than escalating it for glucose control.

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7. Nocturnal diarrhea / enteropathy

WHAT YOU SEE

Nocturnal diarrhea — classic diabetic enteropathy.

WHAT IT ENCODES

Diabetic autonomic enteropathy classically causes painless, watery NOCTURNAL (or explosive postprandial) diarrhea, often alternating with constipation, from dysmotility plus small-intestinal bacterial overgrowth (SIBO). Treat the SIBO component with rifaximin and use loperamide symptomatically. Mnemonic: 'diabetic diarrhea wakes you at night.'

BOARD TRAP

WRONG to anchor on a malabsorptive cause like celiac disease for every diabetic with diarrhea — although celiac clusters with T1DM, the classic boards answer for nocturnal watery diarrhea in long-standing diabetes is autonomic enteropathy ± SIBO. Discriminator: nocturnal timing and dysmotility point to enteropathy; weight loss + iron-deficiency anemia + positive tTG-IgA point to celiac.

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8. Bladder atony

WHAT YOU SEE

Bladder atony: overflow incontinence, UTI, incomplete emptying.

WHAT IT ENCODES

Autonomic neuropathy → neurogenic (atonic) bladder: impaired detrusor sensation and contraction cause incomplete emptying, high post-void residual, urinary retention, overflow incontinence, and recurrent UTIs. Confirm with bladder ultrasound/post-void residual; manage with scheduled (timed) voiding, double voiding, and clean intermittent catheterization for large residuals. Mnemonic: 'the bladder can't feel full and can't squeeze empty.'

BOARD TRAP

WRONG to default to an anticholinergic (oxybutynin) when a diabetic has 'incontinence' — overflow incontinence from an atonic, under-emptying bladder is WORSENED by anticholinergics. Discriminator: check the post-void residual first; a large residual means overflow/retention (needs emptying strategies), not overactive bladder.

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9. Erectile dysfunction / PDE-5

WHAT YOU SEE

Erectile dysfunction: most common GU autonomic manifestation, increases with duration; PDE-5 inhibitors (sildenafil/tadalafil) less responsive than non-diabetics.

WHAT IT ENCODES

Erectile dysfunction is the most common male GU autonomic complication, from combined autonomic neuropathy and endothelial/vascular disease; prevalence rises with diabetes duration and poor control. ED is also an early marker of generalized atherosclerosis (consider CVD risk). First-line therapy is a PDE-5 inhibitor (sildenafil, tadalafil), though response rates are lower than in non-diabetics.

BOARD TRAP

PDE-5 inhibitors are the WRONG choice when the patient uses nitrates — the combination causes life-threatening hypotension (absolute contraindication). Discriminator: screen for nitrate use; if contraindicated, options are intracavernosal alprostadil, vacuum devices, or treating low testosterone if present.

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10. Sudomotor — upper limb hyperhidrosis

WHAT YOU SEE

Upper limbs hyperhidrosis; lower limbs anhidrosis, dry cracked skin, foot ulcers.

WHAT IT ENCODES

Sudomotor (sweat gland) autonomic dysfunction produces a length-dependent pattern: distal/lower-limb ANHIDROSIS with compensatory upper-body/truncal HYPERHIDROSIS, plus gustatory sweating (facial sweating while eating). The dry, anhidrotic lower legs and feet crack and fissure, creating entry points for infection and ulceration. Mnemonic: 'dry feet below, sweaty above.'

BOARD TRAP

WRONG to attribute the dry, fissured diabetic foot to a primary dermatologic/fungal cause and treat only topically — the underlying driver is autonomic anhidrosis. Discriminator: the combination of distal anhidrosis with proximal compensatory sweating points to sudomotor neuropathy; management is emollients plus foot protection, not just antifungals.

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11. Risk factors board

WHAT YOU SEE

Risk factors: smoking, peripheral vascular disease, PAD, foot deformity, nephropathy/retinopathy.

WHAT IT ENCODES

Foot ulcer and amputation risk factors: peripheral neuropathy (loss of protective sensation), PAD, prior ulcer/amputation, foot deformity (Charcot, claw/hammer toes), smoking, poor glycemic control, and coexisting nephropathy/retinopathy (markers of advanced microvascular disease). Annual comprehensive foot exam is required; high-risk feet need podiatry and protective footwear. Mnemonic: 'numb + no blood + deformed = ulcer.'

BOARD TRAP

FEMALE sex is the WRONG answer when asked for a foot-ulcer/amputation risk factor — MALE sex is associated with higher risk. Discriminator: the strongest predictors are loss of protective sensation, PAD, and a PRIOR ulcer (the single best predictor of a future ulcer), not female sex.

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12. Monofilament test

WHAT YOU SEE

Monofilament test: screens for loss of protective sensation; probes to bone -> osteomyelitis likely.

WHAT IT ENCODES

The 10-gram (5.07 Semmes-Weinstein) monofilament screens for loss of protective sensation — the strongest predictor of future ulceration; combine with a 128-Hz tuning fork for vibration. Inability to feel the monofilament at standard plantar sites identifies the at-risk foot. In an existing ulcer, a positive probe-to-bone test markedly raises the likelihood of osteomyelitis. Mnemonic: 'can't feel the wire = can't feel the wound.'

BOARD TRAP

WRONG to rely on the patient REPORTING foot symptoms to detect neuropathy — most loss of protective sensation is asymptomatic, so screening with the monofilament is essential. Discriminator for osteomyelitis: a positive probe-to-bone or an ulcer >2 cm/long-standing should prompt MRI and bone culture, not just empiric antibiotics.

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13. Topical antibiotics limited value

WHAT YOU SEE

Topical antibiotics / antiseptics: limited value, toxic to healing tissue.

WHAT IT ENCODES

Topical antibiotics and antiseptics (e.g., povidone-iodine, hydrogen peroxide) have LIMITED value in diabetic foot wounds and can be cytotoxic to fibroblasts/granulation tissue, impairing healing. Cornerstones of management are sharp DEBRIDEMENT, OFFLOADING (total-contact cast for plantar ulcers), revascularization if PAD, and SYSTEMIC antibiotics for clinical infection. Mnemonic: 'debride, offload, and dose systemically — not paint it.'

BOARD TRAP

Topical antiseptics/antibiotics are the WRONG primary treatment for an infected diabetic foot ulcer — they do not penetrate to deep tissue/bone. Discriminator: a clinically infected ulcer needs systemic antibiotics guided by deep-tissue/bone culture (avoid superficial swabs) plus debridement and offloading.